

Visiobo: Assisting Poster Reading for Multiple Viewers with a Projector-enhanced Physical Agent

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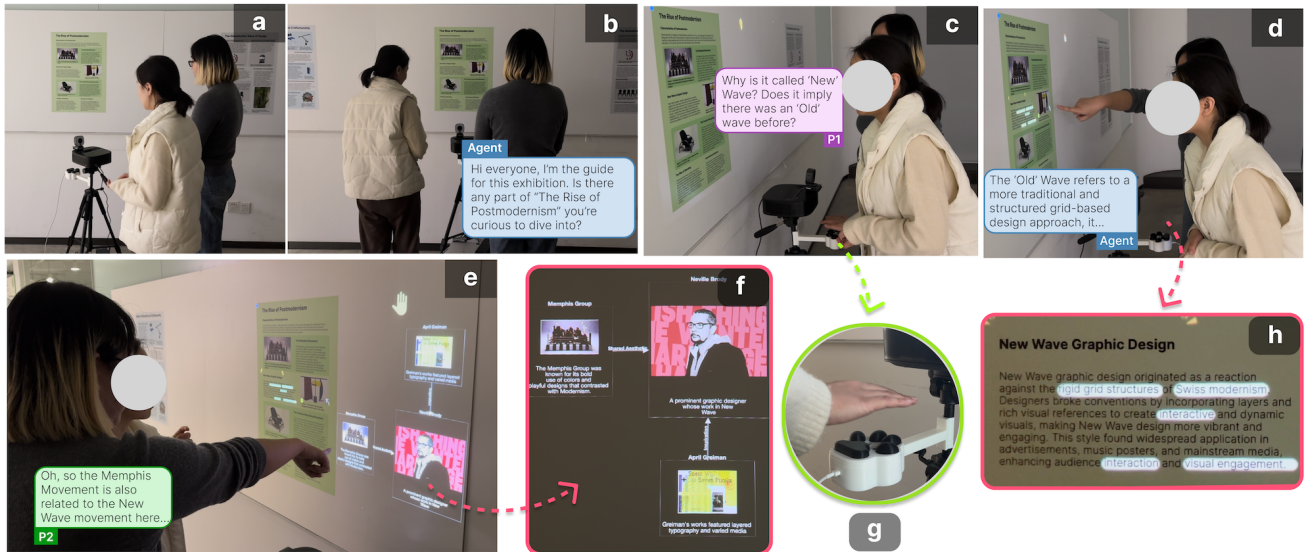


Figure 1: Overview of the interaction with Visiobo. (a) Users are guided to a specific poster by Visiobo. (b) Visiobo introduces the poster's theme and invites users to interact. (c) User 1 asks Visiobo a question, initiating a voice-based interaction. (d) Visual cues such as highlights, arrows, and blinking effects are overlaid on the poster. (e) Expanded information is organized in the form of a concept map around the poster. (f) Details of concept map. (g) A non-contact switch to trigger voice interaction. (h) Details of visual cues.

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Abstract

As an important medium for knowledge dissemination, posters support autonomous learning in public settings. However, static posters with limited space lack interactivity and fail to meet the need for personalized expanded learning. Conversational agents with knowledge bases can offer personalized content, but voice interaction in text-heavy posters may face limitations such as reduced immersion and difficulties in locating information. Research shows that visual information enhancement can reduce cognitive load, yet there is a lack of exploration combining conversational agents with

visual information enhancement. We propose an embodied guide agent for poster exhibitions that supports multi-user interaction through voice and information enhancement (cueing and concept mapping). In an evaluation involving 6 groups with 12 users, results show that the system enables personalized interaction, supports information location and confirmation, provides expanded content, fosters shared understanding among viewers, and enhances the learning experience.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; **User interface design**; • **Computing methodologies** → **Mixed / augmented reality**.

Keywords

Interactive posters, Visual enhancements, Conversational agents, Cueing

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1 INTRODUCTION

Posters play a significant role in facilitating knowledge transfer and exchange. Posters employ multimodal elements such as text, images, and colors to effectively convey key information in an intuitive manner, while supporting viewers in autonomously exploring topics of interest based on their preferences and initiative within public and dynamic environments[8, 20]. However, a poster must be understandable on its own and carry most of the “talking”, yet static posters cannot interact with viewers through questions and answers[29, 34]. Knowledge dissemination posters are often text-dense, making it difficult to quickly extract key information. Constrained by space and a singular organizational format, a single poster cannot meet the diverse needs of readers for expanded knowledge and varied forms of knowledge presentation.

Using conversational agents to enhance learning materials can achieve interactive learning[23], by integrating additional knowledge bases, they can provide personalized expanded knowledge[37]. However, unimodal voice-based conversational agents may lack immersion[7]; moreover, interacting with voice agents while handling text-dense learning materials like posters may contribute to cognitive load[16, 21]. The ephemeral nature of audio makes it difficult for viewers to quickly locate which part of the poster is being referenced in the current interaction and to retain the dialogue content[11, 12]. It has been proved that real-time and interactive visual cues can help learners promptly locate key information, improving knowledge retention and transfer[39]. Nevertheless, HCI research remains scarce in augmenting physical learning materials through the integration of conversational agents and in-situ visual information.

We propose an embodied guide agent, Visiobo, to support the learning experience in poster exhibitions through dialogue-based

interactions and dynamic visual information enhancements. Visiobo is primarily designed to function in situations where poster presenters are unavailable. Rather than replacing a human presenter, it is built to enhance the autonomous viewing experience by offering interactive features, personalized content, and deeper insights. Visiobo interacts with viewers via voice, inviting them in asking and answering questions about the poster content based on their individual interests, and generates dynamic visual information enhancements through real-time reasoning based on the personalized dialogue. The information enhancements consist of two forms: visual cues overlaid on the poster to locate and confirm key information from the dialogue, and concept mapping beside the poster to organize and expand knowledge related to the poster. By openly presenting voice and visual information, Visiobo aims to enable collaborative knowledge acquisition and exchange among multiple viewers.

We evaluated Visiobo with 6 groups comprising 12 participants to explore its capability in supporting users with key information location and confirmation during interactions with a voice agent, providing personalized knowledge expansion, and facilitating multi-user collaborative learning. The within-group experiments were conducted to assess three experimental conditions: (1) voice only, (2) voice with visual cues, and (3) voice combined with visual cues and concept mapping. Qualitative and quantitative results indicated that our system excelled in providing expanded information, supporting collaborative learning, and improving user experience. This study provides valuable insights for designing embodied agents that integrate voice and visual information to support interactive learning with physical learning materials.

2 RELATED WORK

2.1 Conversational Agents for Learning Materials

Conversational agents can enhance the interactivity of learning materials and increase learning engagement. Through voice or text[26, 35], conversational agents can interact with users around electronic books[19] or physical books[40], educational videos, online learning platform resources, museum artifacts and other physical or digital learning materials[3]. For instance, studies have shown that teachable conversational agents, interacting with users via text boxes and guiding them to teach algorithm problems through question-and-answer, can effectively support users’ knowledge construction[17]. Additionally, smart voice assistants have been used to support children’s interactive reading with physical books by enhancing engagement to improve their comprehension[40]. However, unimodal audio-based conversational agents may lack immersion[7]. And voice interactions without spatial alignment make it difficult for users to connect auditory information with visual content[11]. Glaser et al.[12] suggest that audio text design could follow the “Location, Description, Interpretation”(LDI) model by using cues to locate content, followed by description and explanation. However, research on how conversational agents localize information in physical learning materials remains limited.

2.2 Visual information enhancement of learning materials

Enhancing learning materials with real-time visual information can help learners reduce cognitive load[13], optimize knowledge structures[15, 24], and gain supplementary knowledge[10]. Such enhancements include non-content information like visual cues[39], and content-specific information. For example, cueing elements like arrows, highlights, and color coding can help learners locate key information in multimedia courseware[22, 38]. Tabone[33] studied enhancing museum experiences using AR glasses for voice interaction and generating textual information alongside exhibits. However, AR content is only visible to the wearer, making it limited for multi-person collaborative public learning. Projectors can publicly perform visual information enhancement on physical learning materials such as books, maps, or circuits[31, 36]. For example, Schöning et al. [30] proposed combining a handheld projector with electronic map data to project highlights of landmarks, streets, and areas onto paper maps. Ro et al. [27] proposed a mobile projection-based AR robot, designed for museums and galleries to guide visitors and project exhibit information on nearby surfaces. However, its projection information is rule-based rather than generated through real-time reasoning based on user interaction. Current projection-based technologies have not been sufficiently explored in dynamically generating context-aware and personalized information enhancements that align with users' learning backgrounds and individual interests.

3 METHODOLOGY

3.1 Design

3.1.1 Usage Scenarios. To address the challenges faced in current poster reading experiences, we propose an embodied guide agent for poster exhibitions, named Visiobo, which supports multi-viewer interaction. When users enter the exhibition area, Visiobo actively approaches them and inquires about their interests through voice interaction, subsequently guiding them to the corresponding poster (Figure 1-a). Upon arrival at a poster, Visiobo proactively introduces the poster's topic and invites users to ask questions about its content (Figure 1-b). By placing the palm over a non-contact switch (Figure 1-g), users can pose inquiries about the poster's details or explore related knowledge based on their interests (Figure 1-c). After receiving a question, Visiobo first performs real-time reasoning to locate the information within the poster relevant to the user's query (Figure 1-d). Using a projector, it overlays cueing elements such as highlights, blinking effects, or arrows on key components such as keywords, subtitles, or images. These cueing elements provide users with visual guidance and confirmation of the content to be discussed. It then answers the user's question via voice. After that, based on the current dialogue and its knowledge base, it generates extended information in the form of a Concept Mapping adjacent to the poster (Figure 1-e). At the end of the interaction, Visiobo prompts the user with a further question to encourage continued engagement. When nearby users overhear or notice the discussion between the system and the current user and find the topic relevant to their interests, they may be attracted and spontaneously join the conversation.

3.1.2 Key Design Features. As illustrated in the usage scenarios, Visiobo incorporates the following three key design elements: **DF1: A voice-based conversational agent with poster knowledge and expanded knowledge.** Users interact with the conversational agent via voice, where the agent generates responses related to the poster content based on users' inquiries and retrieves relevant information from an extended knowledge base. At the end of each interaction, the agent formulates a follow-up question based on the ongoing conversation and poster content to encourage user reflection and facilitate further engagement. Through questions and answers, Visiobo captures users' attention and guides them to explore and learn specific content based on their interests. Given the time-constrained nature of poster reading, the agent's responses are limited to a maximum of 200 Chinese characters per interaction. **DF2: Cueing overlaid on poster information.** Before answering a user's question, Visiobo uses overlaid cueing elements on the poster to intuitively and efficiently convey key information, helping users clearly identify the focus of their interaction with the conversational agent. The cueing employs three distinct overlay elements: blinking to highlight subtitles and help users quickly locate the current discussion area; highlighting key phrases to visually emphasize the main points of the response; and symbols like arrows to enhance image displays, establishing connections between text and visuals (Figure 1-h). **DF3: Expanded Text and Images Organized in Concept Mapping.** After answering a question, the system generates a concept map near the poster, presenting expanded information related to the poster's content through concise text and images. Arrows are used to illustrate the connections between different pieces of information. Balancing the depth of additional information with the short-duration reading characteristics of posters, we limit the number of expanded knowledge nodes to 3–4. The system not only provides additional knowledge tailored to the needs of different users but also introduces a alternative way of organizing knowledge based on the specific content each user focuses on (Figure 1-f).

3.1.3 System Workflow and Implementation. Visiobo's workflow consists of two phases: preprocessing and real-time interaction (Figure 2-a). In the preprocessing phase, Visiobo identifies poster text coordinates and builds a knowledge base. Using open-source OCR technology[9], it detects precise word-level coordinates. Through spatial clustering with the k-means algorithm and semantic analysis by GPT-4o, the poster content is segmented into distinct regions. Meanwhile, the content of *History of Design* is vectorized to form the knowledge base. In the real-time interaction phase, Visiobo operates on a multi-agent framework powered by GPT-4o, comprising a Cueing and Conversation Agent and a Concept Mapping Agent. The Cueing and Conversation Agent converts user voice input into text via Whisper[2], performs real-time reasoning to generate cueing content and corresponding replies, which are voiced using CosyVoice[1]. The conversation is passed to the Concept Mapping Agent, which uses RAG technology to infer related knowledge and relationships from the knowledge base and generates a relational graph with the fCoSE layout algorithm[4].

The hardware system's key components are shown in Figure 2-b. It consists of a projector for audio output and visual enhancements, a portable power bank for powering the projector, an ultrasonic

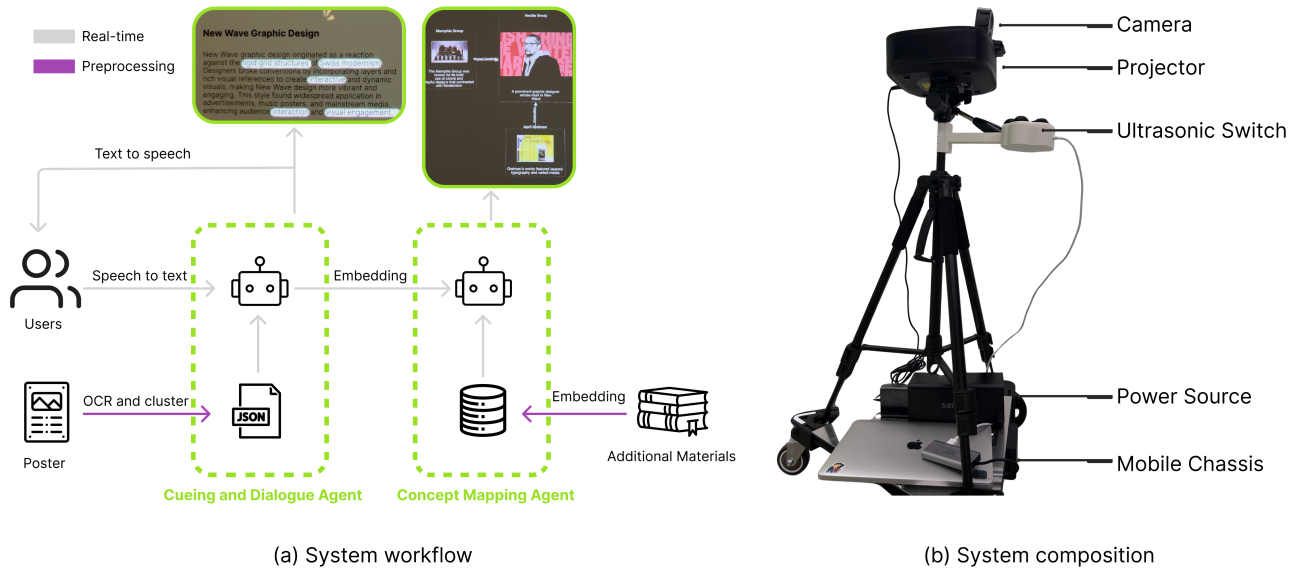


Figure 2: System workflow and system composition.

sensor for triggering voice input reception, a laptop for real-time speech input processing and generating dialogue and visual enhancements, a webcam, and a mobile intelligent chassis for aligning the projector with the poster.

Our consideration to utilize a mobile chassis rather than fixed projectors was driven by practical considerations for real-world deployment and future scalability. In real poster exhibitions with numerous displays, equipping each poster with its own projector would be expensive and complex to install. The mobile platform allows a single system to serve multiple posters throughout an exhibition space, reducing hardware requirements while maintaining functionality. Additionally, the mobility enables the system to actively approach and guide visitors between posters based on their expressed interests, leading to a more dynamic and personalized exhibition experience than stationary installations could provide. mobility enables a single agent to maintain the interaction context between the user and multiple posters, ensuring a more cohesive and immersive interactive experience.

3.2 User Study

3.2.1 Participants. We recruited 12 participants (7 males, 5 females) from college students and staff from the researchers’ institute, aged 18 to 44, with professional backgrounds in design, engineering, and education. The participants were divided into six groups, each consisting of two individuals, with three groups comprising participants who were familiar with each other and the other three groups comprising those who were unfamiliar with each other. A questionnaire was used to collect information on the participants’ familiarity with the experimental poster topic, their usage of voice-based AI products, their frequency of viewing posters, and demographic information (see Appendix A for details). All participants completed

an informed consent procedure prior to participation. For clarity, we labeled the groups as G1 to G6 and the participants as P1 to P12.

3.2.2 Procedure. We conducted a within-subject experiment to compare three experimental conditions: Condition 1 (Voice), Condition 2 (Voice + Cueing), and Condition 3 (Voice + Cueing + Concept Mapping). The sequence of the three conditions was fully counterbalanced, with a minimum interval of two hours between each condition. Based on the book *A History of Design*, we created three posters on different topics related to design history. The posters combined text and visuals to showcase design works, figures, movements, and styles from various periods, and a mini poster exhibition environment was set up (see Figure 1).

Each evaluation session included the following activities: **Introduction and training (5 minutes)** — Before the first experiment in each group, the researchers introduced the system’s usage scenarios and interaction methods, guiding participants to familiarize themselves with the system interaction beforehand. **Experiment (16 minutes)** — This stage was divided into two phases: “taking turns asking the system questions” and “freely interacting with the system,” each lasting approximately 8 minutes. During both phases, participants were free to communicate and discuss with each other. **Scale evaluation (10 minutes)** — After each condition, participants completed the User Engagement Scale (UES)[18], the System Usability Scale (SUS)[25], and a customized six-item scale (A1-A6). The customized scale explored the system’s performance in helping participants locate and confirm information on the poster, providing personalized supplementary information, and supporting collaborative learning (see Appendix B for details). **Interviews with each group (15 minutes)** — After completing all three experimental conditions, participants were invited to join focus group interviews to provide feedback on their experiences

with the system’s design elements and its support for collaborative learning.

4 RESULTS

We report the quantitative results of the three scales and provide representative examples from participants interview statements to support the findings. When evaluating the quantitative data under the three experimental conditions, we first used the Shapiro-Wilk test to assess the normality of the data. The results showed that the scores of the System Usability Scale (SUS) and the User Engagement Scale (UES) followed a normal distribution ($p > 0.05$). Accordingly, we conducted a repeated-measures one-way analysis of variance (RM-ANOVA) and performed Tukey’s HSD post-hoc test. For the custom scale, A1&A2, A3&A4, and A5&A6 did not meet the normality assumption ($p < 0.05$). Therefore, we applied the Friedman test and conducted Conover post-hoc comparisons. The overall results are shown as boxplots in Figure 3.

Through quantitative analysis, the data revealed significant differences in the following aspects: (1) **User engagement**: Condition 3 (Voice + Cueing + Concept Mapping) demonstrated a significant improvement over Condition 1 (Voice) ($p = 0.034$). (2) **Personalized and extended information**: Condition 3 (Voice + Cueing + Concept Mapping) also showed a notable improvement over Condition 1 (Voice) ($p < 0.001$). (3) **Collaborative understanding**: Condition 2 (Voice + Cueing) significantly enhanced collaborative understanding of the poster content compared to Condition 1 (Voice) ($p = 0.009$). Condition 3 (Voice + Cueing + Concept Mapping) further improved collaborative understanding compared to Condition 1 ($p < 0.001$). The following sections integrate user interview data to provide a detailed interpretation of each aspect.

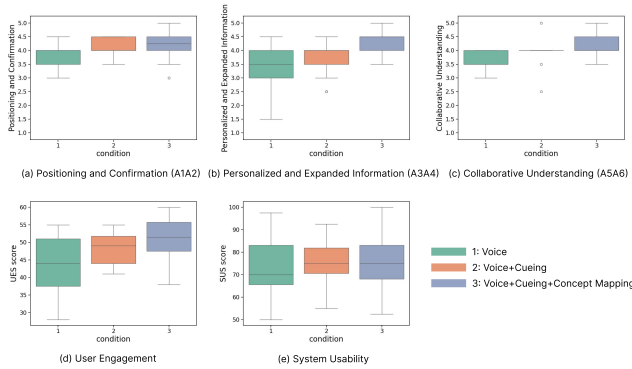


Figure 3: The boxplots illustrate user ratings across three conditions (Condition 1: Voice, Condition 2: Voice+Cueing, Condition 3: Voice+Cueing+Concept Mapping), showing median values, interquartile ranges, and variability.

Information Positioning and Confirmation during Conversation. For A1&A2, “It is easy for me to confirm whether the voice agent understands my question” and “I can clearly identify which part of the poster the voice agent is referring to during the discussion,” there was no significant effect of condition ($p = 0.336$). Nevertheless, qualitative interview data offered some evidence of

its effects. For instance, because cueing occurs before the voice agent speaks, the visual cues act as a buffer, allowing users to prepare for the system’s response and quickly grasp the key points. As G1P2 mentioned, “*Cueing is very helpful. I can quickly grasp the key points, especially because the poster content is very dense.*” Additionally, participants noted that cueing reduced uncertainty during their interactions with the system. G1P1 expressed, “*As soon as the cueing appeared, I knew what the agent was going to talk about.*”

Personalized and Expanded Information. For A3&A4, “The system provides information based on the topics I am interested in” and “The system provides information that is not included on the poster,” a significant difference was observed between Condition 1 (Voice) and Condition 3 (Voice + Cueing + Concept Mapping) ($p < 0.001$). However, no significant differences were found between Condition 1 (Voice) and Condition 2 (Voice + Cueing) ($p = 0.082$) or between Condition 2 (Voice + Cueing) and Condition 3 (Voice + Cueing + Concept Mapping) ($p = 0.082$). This suggests that the Concept Mapping generated based on users’ questions provided personalized and expanded information. G1P2 noted, “*I wanted to understand the cause and effect of the events mentioned on the poster, and the concept map was able to answer my questions.*” G6P12 added, “*The representative design works in the concept map were particularly impressive.*” Similarly, G5P10 mentioned that the concept map effectively enhanced their understanding.

Collaborative Understanding. For A5&A6, “I had meaningful discussions with my partner about specific topics related to the poster” and “I could clearly understand which part of the poster my partner was focusing on,” significant differences were observed between Condition 1 (Voice) and Condition 2 (Voice + Cueing) ($p = 0.009$) as well as between Condition 1 (Voice) and Condition 3 (Voice + Cueing + Concept Mapping) ($p < 0.001$). However, no significant differences were found between Condition 2 (Voice + Cueing) and Condition 3 (Voice + Cueing + Concept Mapping) ($p = 0.087$). This indicates that cueing visually guided participants’ shared attention. As G2P4 mentioned, “*Cueing helped me understand that the topic we were discussing was focused on a specific area, which I found very useful.*” Additionally, Concept Mapping served as a medium to facilitate collaborative understanding. G2P3 remarked, “*The poster mentioned Philippe Starck. I knew he had a famous work, the juicer, but my partner didn’t. When we asked the system, it displayed the juicer’s image on the concept map, which allowed us to have a better discussion.*”

User engagement. Figure 3-d illustrates a progressive increase in user engagement across the three conditions. A significant difference was observed between Condition 1 (Voice) and Condition 3 (Voice + Cueing + Concept Mapping) ($p = 0.034$), indicating that the inclusion of visual enhancements such as cueing and concept mapping significantly boosted user engagement compared to the voice-only condition.

From the interviews, participants expressed that the dialogue-based interaction was helpful but somewhat passive, requiring more active engagement with the content to sustain interest. However, cueing and concept mapping attracted their attention in complementary ways. Participants noted that cueing effectively captured their focus and deepened their understanding of the poster content. For example, G3P5 remarked, “*I immediately understood the key points of this section,*” while G5P9 compared it to the use of

highlighted words in a PowerPoint presentation: “It’s like the highlighted keywords in a PPT.” As for concept mapping, participants found it added novelty and enjoyment to their experience. G6P12 mentioned, “It supplemented the poster with many images I hadn’t seen before, which left a strong impression,” while G2P3 expressed, “It provided additional knowledge, which made me more eager to explore further.”

System Usability. The mean System Usability Scale (SUS) scores for all three conditions exceeded 70 (out of 100), indicating that all systems performed well in terms of usability.

5 DISCUSSION

5.1 Shared Audiovisual Information Facilitated Joint Understanding

Posters support the public dissemination of knowledge. However, discussions around posters primarily occur between the presenter and the viewers, and in the absence of the presenter, posters struggle to facilitate knowledge exchange effectively[28]. Different viewers may hold diverse and interesting perspectives due to their varied background knowledge and differing approaches to interpreting the poster content. Social Learning Theory highlights that observing others’ behaviors or focal points is a crucial way for individuals to learn[5]. However, this diversity of individual focus often remains unnoticed by others in the absence of a sharing mechanism[6]. When viewers engage independently or are faced with large amounts of information, they often fail to perceive the points of interest or discussions of others, resulting in a lack of collective engagement[14, 32].

Our system addresses this issue by encouraging viewers to ask questions and respond to others, thereby fostering the expression of individual interests and personal understanding of the poster content. This, in turn, provides new perspectives for other audience members. Through cueing, viewers can intuitively observe the focal points of others’ attention, establishing shared focus. Additionally, concept mapping visualizes the knowledge elements of interest to different audience members, promoting a collective understanding of the content. In our experiments, even unfamiliar groups of viewers, without direct communication, could gain inspiration by observing and listening to each other’s interactions with the poster. This demonstrates that our system can indirectly facilitate shared understanding and knowledge exchange between strangers in public settings, thereby enhancing the learning experience. Beyond poster exhibition scenarios, our system also holds potential applications in other public learning environments with diverse audience knowledge backgrounds, such as museums and tourist attractions.

5.2 Limitations and Future Work

One limitation of this study is that current experiments were conducted in a laboratory environment, where we could precisely control participant positioning, number and lighting conditions. While this approach allowed us to effectively evaluate the core functionality of the system, future work should explore performance in authentic exhibition environments where lighting conditions vary and visitor movement is unpredictable. Additionally, future work should also consider how the mobile hassis can navigate safely in a multi-user poster reading environment. Another limitation

is that while our system successfully provided expanded content through concept mapping, we did not directly measure the cognitive load that this additional information placed on users. The balance between providing enriching supplementary information and overwhelming viewers remains an important consideration.

6 CONCLUSION

We designed an embodied guide agent to support interactive and collaborative poster reading experiences in exhibition scenarios. The system engages users through personalized interactions with a conversational agent and employs visual enhancements to guide users’ attention to poster content while providing additional extended information. A user study involving 12 participants, divided into 6 groups, evaluated three design features of the system (voice, cueing, and concept mapping). The results demonstrated that the system effectively supported users in locating and identifying poster content, offering personalized supplementary information, and fostering a shared understanding of the poster content. Our study highlights the potential of combining conversational agents with visual information enhancements to improve the interactivity of learning materials and facilitate collaborative learning, providing valuable insights for the design of such embodied agents.

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A Participants Information

Table 1: Participants Information

Category	Details
Gender	Male (7), Female (5)
Age	18–24 (6), 25–35 (5), 35–44 (1)
Field of Study	Art and Design (4), Science and Engineering (6), Education (2)
Familiarity with the Poster Topic	Completely unfamiliar (7), Basic understanding (3), In-depth understanding (2)
Frequency of Viewing Posters	Yearly (1), Monthly (6), Weekly (3), Daily (2)
Frequency of Using Voice-based AI	Never (1), Monthly (4), Weekly (4), Daily (3)

B Custom Questionnaire

A1: It is easy for me to confirm whether the voice agent understands my question.

A2: I can clearly identify which part of the poster the voice agent is referring to during the discussion.

A3: The system provides information based on the topics I am interested in.

A4: The system provides information that is not included on the poster.

A5: I had meaningful discussions with my partner about specific topics related to the poster.

A6: I could clearly understand which part of the poster my partner was focusing on.

Each statement has five response options: Strongly disagree, Disagree, Neither agree nor disagree, Agree, Strongly agree.